

AWS LOAD BALANCERS

Complete Practical Guide

From networking fundamentals to production design, security, observability, cost control and troubleshooting

ALB	NLB	GWLB	CLB
HTTP-aware routing	High-performance flows	Network appliances	Legacy migration

Structured learning path | Architecture patterns | Hands-on labs | Troubleshooting
playbook | Interview review

How to use this guide

This guide teaches Elastic Load Balancing as a system, not as a list of console settings. Read Parts I through III in order, then use the remaining parts according to your role. Every chapter ends with a compact recap or review prompt.

If you are...	Prioritize	Outcome
New to AWS	Parts I, II and III	Understand traffic flow and choose the correct load balancer
Application engineer	ALB, security, ECS/EKS and troubleshooting	Deploy web services safely
Network engineer	NLB, GWLB, routing and resilience	Design predictable packet and flow paths
Cloud/DevOps engineer	Operations, cost, IaC and labs	Build and operate production platforms
Preparing for interviews	Decision matrix, recaps, glossary and review questions	Explain tradeoffs clearly

Learning rule

For every feature, answer four questions: What problem does it solve? Where is it configured? What changes in the traffic path? What failure mode does it introduce?

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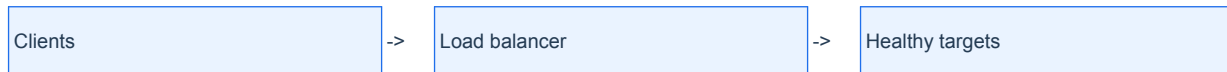
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PART I - FOUNDATIONS

1. Why Load Balancers Exist

A load balancer is a controlled entry point that accepts traffic and forwards it to one of several destinations. The important word is controlled: the service applies health, routing, security and lifecycle decisions before a target receives traffic.



Problems solved

Problem	Without a load balancer	With Elastic Load Balancing
Server failure	Clients continue calling a dead address	Unhealthy targets are removed from normal routing
Traffic growth	One server becomes a bottleneck	Targets can be added behind a stable endpoint
Deployments	Clients must learn new server addresses	Targets register, warm up and drain behind the endpoint
Security	Every target needs direct exposure	Targets can remain private while the load balancer is public
Routing	Applications implement edge-routing logic	ALB can route by HTTP request content

Load balancing is not the same as scaling

The load balancer distributes traffic, but it does not automatically create application capacity. An Auto Scaling group, ECS service or Kubernetes controller adjusts target capacity. ELB then discovers or receives registrations for those targets.

Key distinction

High availability means the service can survive component failure. Scalability means it can handle changing demand. A design needs both; neither automatically guarantees the other.

Chapter recap

- ELB is a managed traffic entry point
- Health and routing decisions happen before a target receives traffic
- Scaling systems add capacity; ELB distributes traffic to it

PART I - FOUNDATIONS

2. Networking Foundations

IP addresses, ports and protocols

An IP address identifies a network interface. A port identifies a listening application on that interface. A protocol defines how the endpoints exchange data. A security rule that permits an IP but not the required port still blocks the application.

Concept	Example	Why ELB cares
IP address	10.0.2.15	Identifies nodes and targets
Port	443	A listener accepts traffic on a configured port
TCP	Reliable connection	NLB routes TCP flows; HTTP commonly uses TCP
UDP	Connectionless datagrams	NLB can balance UDP workloads
TLS	Encrypted session	ALB/NLB can terminate encryption
HTTP	Request/response protocol	ALB reads host, path, headers and method

Layers 3, 4 and 7

The layer determines what information the load balancer can use. A lower-layer device sees less application meaning but can support more protocols. A higher-layer device understands the application protocol and can make richer decisions.

Layer	Information used	AWS service	Typical decision
3 - Network	IP packets	GWLB	Send traffic through a firewall appliance
4 - Transport	IP, port and protocol	NLB	Map a flow to a target
7 - Application	HTTP request content	ALB	Route /orders to the orders service

DNS is part of the availability design

Clients normally reach an ELB through its AWS DNS name or a Route 53 alias. DNS returns load-balancer addresses, and clients choose an address. Because these addresses can change for managed services, applications should use DNS names instead of storing resolved ALB addresses.

Common mistake

Do not copy an ALB's currently resolved IP addresses into an allowlist and assume they are permanent. Use an NLB with static zonal addresses when a fixed-address requirement is real.

Check your understanding

- 1 Why can an ALB route by URL path while an NLB normally cannot?
- 2 What is the difference between an IP address and a port?
- 3 Why should clients use the ELB DNS name?

PART I - FOUNDATIONS

3. How Elastic Load Balancing Works



Load-balancer nodes and Availability Zones

When a zone is enabled, ELB places managed capacity in the selected subnet for that zone. Targets registered in a zone do not receive traffic through the load balancer unless the zone is enabled. Production designs normally enable multiple zones and maintain healthy target capacity in each.

The two connection model

ALB acts as a reverse proxy. It maintains a front-end connection with the client and a separate back-end connection with the target. This allows TLS termination, header processing and target selection, but it also means front-end and back-end timeouts must be understood separately.

```
Client == front-end connection ==> ALB == back-end connection ==> Target
```

Core resource hierarchy

Resource	Question it answers	Example
Load balancer	Where does traffic enter?	Internet-facing ALB in three subnets
Listener	Which protocol and port are accepted?	HTTPS on 443
Rule	What should happen for this request?	Path /api/* forwards to API group
Target group	Which pool and health policy apply?	API targets on port 8080
Target	Which destination handles the work?	ECS task at 10.0.4.22:8080

Mental model

The listener accepts. The rule decides. The target group defines the pool and policy. The target performs the work.

PART II - LOAD BALANCER TYPES

4. Choosing the Correct Load Balancer

Type	Layer	Best fit	Signature capability
ALB	7	Websites, APIs, microservices	Content-based HTTP routing
NLB	4	High-throughput or non-HTTP traffic	Static zonal IPs and client IP preservation
GWLB	3	Virtual network appliances	Transparent service insertion with GENEVE
CLB	4/7	Existing legacy deployments	Backward compatibility, not new design

Decision sequence

- 1 Does routing depend on HTTP host, path, header, query string, method or authentication? Choose ALB.
- 2 Do you require TCP, UDP, QUIC, static IPs, very high flow performance or PrivateLink? Choose NLB.
- 3 Are you inserting firewalls, IDS/IPS or deep packet inspection appliances? Choose GWLB.
- 4 Is it an existing Classic Load Balancer? Operate it safely, then plan migration.

Services can be combined

Selection is not always exclusive. CloudFront may sit in front of an ALB. Global Accelerator can provide global static anycast addresses in front of regional ALBs or NLBs. An NLB can target an ALB when static entry points or PrivateLink must be combined with Layer 7 routing.

Exam and interview shortcut

Start with the protocol and routing requirement. Then evaluate address stability, client IP, security integrations, scale characteristics and cost.

PART II - LOAD BALANCER TYPES

5. Application Load Balancer Deep Dive

ALB is an HTTP-aware reverse proxy. It understands the request and can make a different decision for two requests arriving on the same IP address and port.

Protocols and application features

- HTTP and HTTPS listeners, including HTTP/2 on supported front-end connections.
- WebSocket upgrades for long-lived bidirectional connections.
- HTTP/2 and gRPC target-group protocol versions for compatible backends.
- TLS termination, multiple certificates through SNI and optional back-end HTTPS.
- Integration with AWS WAF, Cognito, OIDC authentication and JWT validation.

Rule conditions

Condition	Example	Use case
Host	api.example.com	Multiple domains on one ALB
Path	/orders/*	Microservice routing
Header	X-Tenant: gold	Tenant or client segmentation
Method	POST	Separate write workloads
Query string	version=2	Controlled API routing
Source IP	203.0.113.0/24	Network-based routing condition

Actions

Every rule ends with a routing action: forward, redirect or fixed response. HTTPS rules may perform authentication or token validation before routing. Forward actions can use multiple weighted target groups, enabling controlled releases.

```
Priority 10: host=api.example.com AND path=/orders/* -> orders-tg
Priority 20: path=/static/*                          -> static-tg
Default                                           -> web-tg
```

Target types and protocol versions

ALB target groups support instance IDs, private IP addresses and Lambda functions. Instance targets route to the primary private IP of the registered instance. IP targets are useful for ECS tasks, Kubernetes pods, connected VPCs and reachable on-premises systems. HTTP/2 and gRPC require compatible listeners, target types and application behavior.

Design insight

Use one ALB for several related services when shared routing and security policy are desirable. Use separate ALBs when blast radius, ownership, lifecycle or compliance boundaries matter more than consolidation.

PART II - LOAD BALANCER TYPES

6. ALB Routing, Algorithms and Session Behavior

Target selection algorithms

Algorithm	Behavior	Good fit	Caution
Round robin	Rotates requests across healthy targets	Similar request costs	Does not account for work already in progress
Least outstanding	Prefers target with fewer unfinished requests	Variable request duration	Long-lived requests affect the count
Weighted random	Uses target weights	Adaptive distribution	Understand interaction with anomaly mitigation

Stickiness

Without stickiness, each request is independently eligible for any healthy target. With target stickiness, ALB uses a cookie to map a client session to a particular target. Duration-based stickiness uses an ALB-generated cookie. Application-based stickiness follows a named application cookie.

```
No stickiness: user request 1 -> A, request 2 -> C
With stickiness: user request 1 -> A, request 2 -> A
```

Stickiness can preserve local session state, but it can also produce uneven load and complicate deployments. A more scalable pattern is to store session data in ElastiCache, DynamoDB or another shared store so that any target can handle the next request.

Slow start and deregistration delay

Slow start gradually increases the traffic share of a newly healthy target. Deregistration delay gives existing requests time to complete after a target starts draining. Together, they protect both the beginning and end of a target's lifecycle.

Weighted target groups

A forward action can split traffic between target groups. This supports canary and blue-green releases, but weights are not a substitute for monitoring. Define rollback metrics before shifting production traffic.

Release guardrail

Before increasing a canary weight, compare target 5xx, latency, saturation and business-success metrics between versions. A technically healthy target can still return incorrect results.

PART II - LOAD BALANCER TYPES

7. Network Load Balancer Deep Dive

NLB operates on connections and network flows. It does not need to parse ordinary HTTP routing information, which makes it suitable for non-HTTP protocols and very high-throughput connection workloads.

Listener protocols

Current NLB listener families include TCP, TLS, UDP, TCP_UDP, QUIC and TCP_QUIC. Protocol support and regional availability should always be checked before implementation because newer capabilities evolve over time.

Flow mapping

A flow is selected using network attributes such as source address, source port, destination address, destination port and protocol. Once mapped, packets for that flow remain associated with the selected target for the flow lifetime.

Static zonal addresses

NLB creates a network interface and stable address in each enabled Availability Zone. An internet-facing NLB can use one Elastic IP per enabled subnet. This is valuable for partners and firewalls that require address allowlisting.

Client IP and Proxy Protocol v2

Client IP preservation allows compatible targets to observe the original source address. Availability and defaults depend on target type, address family and protocol. Where direct preservation is unsuitable, Proxy Protocol v2 can carry connection metadata to an application that understands the binary header.

Operational warning

Enabling Proxy Protocol on a target that does not understand it can make application traffic and health checks fail. Treat it as an end-to-end protocol change, not a cosmetic setting.

TLS termination and pass-through

Pattern	NLB behavior	When to use
TCP pass-through	Encrypted bytes reach the target unchanged	Target must own TLS and client-certificate logic
TLS termination	NLB decrypts using an attached certificate	Central certificate and policy management
TLS to TLS target	NLB terminates and opens encrypted back-end connection	Encryption on both segments

PART II - LOAD BALANCER TYPES

8. NLB Advanced Patterns

Source-IP stickiness

NLB target stickiness uses source IP. This is useful for stateful servers, but many users behind the same NAT gateway can share one source IP and become concentrated on one target. Target health changes or registrations can also reset mappings.

PrivateLink endpoint services

An NLB can back an endpoint service. Consumers create interface VPC endpoints and reach the service privately through AWS PrivateLink, without VPC peering, transit routing or public internet exposure. The provider and consumer can be in different accounts.

NLB targeting ALB



This pattern combines NLB features such as static entry addresses or PrivateLink with ALB content routing, authentication and WAF. It adds another hop and another billable resource, so use it only when the combined feature set is required.

Cross-zone behavior

NLB cross-zone load balancing is disabled by default. With it disabled, each zonal node routes to targets in its own zone. This preserves zonal independence but requires enough healthy capacity per zone. With it enabled, distribution can be smoother across unequal target counts.

Connection lifecycle

NLB target groups expose controls for deregistration delay, connection termination for unhealthy targets, unhealthy draining behavior and client IP preservation. Their correct combination depends on whether the application values fast failover or continuity of established connections.

- Chapter recap
- NLB makes flow-level decisions
 - Static zonal addresses solve real allowlisting requirements
 - Client IP and Proxy Protocol require end-to-end validation
 - PrivateLink is a major NLB use case

PART II - LOAD BALANCER TYPES

9. Gateway Load Balancer Deep Dive

GWLB distributes traffic across virtual network appliances while keeping inspection transparent to the application. It combines gateway behavior with managed appliance scaling and health-based selection.



Main components

Component	Role
Gateway Load Balancer endpoint	A PrivateLink-powered route-table destination in the consumer VPC
Gateway Load Balancer	Selects a healthy appliance and preserves flow affinity
GENEVE target group	Registers compatible appliances and defines health/lifecycle policy
Virtual appliance	Performs firewall, IDS/IPS, NAT or inspection work

GENEVE and flow symmetry

GWLB encapsulates traffic to appliances using GENEVE on port 6081. Stateful appliances need both directions of a connection to traverse compatible state. Route tables and appliance design must preserve symmetric traffic flow.

Centralized inspection

A common model places appliances in a dedicated inspection VPC. Multiple spoke VPCs use endpoints and route tables to send selected traffic through the fleet. This centralizes policy but makes route ownership, failure behavior and appliance capacity critical architectural concerns.

Failure question

Decide whether traffic should fail open or fail closed when inspection capacity is unavailable. The correct answer depends on security policy, application criticality and regulatory requirements.

PART II - LOAD BALANCER TYPES

10. Classic Load Balancer and Migration

Classic Load Balancer is the previous ELB generation. It can still appear in established environments, but it lacks many modern routing, target and observability capabilities.

Why migrate

- ALB adds host/path/header routing, redirects, fixed responses, richer target types and application integrations.
- NLB adds static zonal addresses, modern flow performance and IP targets.
- Modern target groups allow independent service health and lifecycle policies.

Safe migration sequence

- 1 Inventory listeners, certificates, health checks, stickiness, security groups, DNS and application dependencies.
- 2 Choose ALB for HTTP-aware workloads or NLB for transport-level workloads.
- 3 Build the new load balancer and target groups in parallel.
- 4 Validate headers, client IP behavior, timeouts, health checks and logs.
- 5 Shift DNS or upstream routing gradually while monitoring both environments.
- 6 Retain a rollback window before removing the Classic Load Balancer.

Migration trap

A migration wizard can copy configuration, but it cannot prove application compatibility. Validate behavior under real protocols, headers, long-lived connections and failure conditions.

PART III - SHARED ELB CONCEPTS

11. Schemes, Address Families, Subnets and Targets

Internet-facing versus internal

Scheme	Address reachability	Typical use
Internet-facing	Publicly reachable load-balancer addresses	Public website or API
Internal	Private load-balancer addresses	Service-to-service or corporate application

An internet-facing load balancer can forward to private targets. The targets do not need public addresses because the load balancer reaches them through VPC networking.

IP address types

Depending on load-balancer type, AWS supports IPv4 and dual-stack modes, and certain ALB configurations can avoid public IPv4 while retaining public IPv6. The load-balancer address type and target-group address type are related but separate choices.

Subnet selection

Selecting a subnet enables its Availability Zone. Ensure each selected subnet has enough free addresses for managed ELB capacity and that routing, network ACLs and target reachability are correct. Do not assume that placing an ALB in a public subnet makes its targets public.

Target registration modes

Target type	What is registered	Common use
Instance	EC2 instance ID	Auto Scaling groups
IP	Private IP and port	ECS, EKS, connected networks
Lambda	One Lambda function	ALB-triggered serverless HTTP
ALB	One ALB behind NLB	Static/PrivateLink entry plus Layer 7 routing

PART III - SHARED ELB CONCEPTS

12. Listeners, Rules, Target Groups and Health

Listener design

A listener defines the front-door protocol and port. Listener security policy, certificates and default action belong at this layer. Keep the public listener set minimal; every open listener is another interface to secure and operate.

Rule priority

ALB evaluates rules from the lowest numeric priority toward the default rule. Specific rules should normally precede broad patterns. Overlapping patterns can silently send traffic to an unintended service, so test positive and negative matches.

Target-group boundaries

Create separate target groups when services need different ports, protocols, health checks, deployment lifecycles, algorithms or observability. A target can belong to more than one target group, but each registration should serve a clear purpose.

Health-check anatomy

Setting	Meaning	Bad configuration effect
Protocol/port	How ELB connects	Every target appears unreachable
Path	Application endpoint tested	False unhealthy or shallow test
Interval	Time between checks	Slow detection or excess check load
Timeout	Wait before failure	False failures on slow endpoints
Thresholds	Consecutive results needed	Flapping or slow recovery
Matcher	Successful response codes	Healthy application marked unhealthy

Designing a health endpoint

A liveness endpoint should be fast and deterministic. A readiness endpoint may check whether the process can actually serve traffic. Avoid checking every downstream dependency if one optional dependency would cause the entire fleet to flap together.

Fail-open behavior

When every target in an ALB target group is unhealthy, ALB can route to all registered targets. This avoids a health-check mistake automatically becoming a total outage, but it also means health checks are not an absolute traffic firewall.

PART III - SHARED ELB CONCEPTS

13. Availability, Cross-Zone Routing and Recovery

Multi-AZ architecture

A resilient regional deployment enables multiple zones and maintains enough target capacity to absorb a zone loss. Merely selecting three subnets does not create resilience if all useful targets or databases remain in one zone.

Cross-zone load balancing

Mode	Routing scope	Strength	Risk
Enabled	Each node can use targets in all enabled zones	Smooth distribution across unequal fleets	Cross-zone dependency and possible transfer considerations
Disabled	Each node uses local-zone targets	Zonal independence and locality	Capacity must be balanced per zone

ALB cross-zone routing is enabled at load-balancer level, with target-group controls available for specific behavior. NLB and GWLB are disabled by default. Defaults are starting points, not design answers.

DNS fail-away and unhealthy routing thresholds

Target-group health thresholds can influence whether a zonal load-balancer address remains in DNS and whether routing expands to otherwise unhealthy targets. These controls trade capacity preservation against the risk of sending traffic to degraded targets.

Zonal shift

Amazon Application Recovery Controller can shift an eligible load balancer away from an impaired Availability Zone. A zonal shift is useful only if remaining zones have enough application and dependency capacity.

Multi-region recovery

ELB is regional. Route 53, Global Accelerator or CloudFront may direct traffic between regional stacks. Multi-region recovery also requires data, secrets, certificates, deployment artifacts and operational procedures to exist in the recovery region.

Capacity test

Design the steady state so that the surviving zones can handle peak traffic after one zone is removed. Then test the assumption with controlled failure exercises.

PART III - SHARED ELB CONCEPTS

14. Connections, Timeouts and Target Lifecycle

Connection reuse

Clients may send many HTTP requests over one front-end connection, while ALB can reuse back-end connections to targets. Connection reuse reduces handshake overhead, but mismatched keep-alive settings can create resets and intermittent 502 responses.

Important timers

Timer	Purpose	Typical symptom when wrong
Idle timeout	Closes inactive connections	Long requests or WebSockets disconnect
Client keep-alive	Limits client connection lifetime	Connections persist longer than desired
Application timeout	Limits server processing	Target returns timeout before ALB
Deregistration delay	Drains work during removal	Deployments cut off requests or take too long
Health timeout	Limits each health probe	Targets flap under temporary latency

Graceful deployment sequence

- 1 Stop assigning new work to the target by starting deregistration.
- 2 Allow in-flight work to complete during the draining period.
- 3 Stop the application after the platform's termination grace period.
- 4 Terminate the instance, task or pod only after drain behavior is understood.

Long-running work

Synchronous requests that run for many minutes are fragile behind network timeouts. Prefer an asynchronous job API: accept work, return a job identifier, process in the background and expose status separately.

Chapter recap

- Front-end and back-end connections are separate
- Timeouts across client, load balancer and target must align
- Deregistration delay and platform termination grace must be coordinated

PART IV - SECURITY

15. Security Groups, NACLs and Network Boundaries

Security-group chaining



The load-balancer security group should allow only required listeners from approved sources. The target security group should allow the application and health-check ports from the load-balancer security group where supported. This prevents direct public access to targets.

Security groups versus NACLs

Control	State	Scope	Operational implication
Security group	Stateful	Network interface	Return traffic for allowed flows is automatically permitted
Network ACL	Stateless	Subnet	Both directions and ephemeral ports must be allowed

Internal does not mean automatically trusted

An internal load balancer removes direct internet reachability, but callers inside connected networks may still be untrusted or compromised. Apply authentication, authorization, segmentation and encryption according to data sensitivity.

IAM and resource administration

IAM controls who can create or change ELB resources; it does not filter application packets. Use least-privilege deployment roles, CloudTrail for API audit, resource tags for ownership and change review for listener/security modifications.

Troubleshooting principle

A healthy target proves that the load-balancer nodes can reach the health endpoint. It does not prove that clients can reach the listener or that every application route works.

PART IV - SECURITY

16. TLS, Certificates and Authentication

TLS termination patterns

Pattern	Traffic path	Tradeoff
Edge termination	Client --TLS--> LB --plain--> target	Simple target configuration; private segment is unencrypted
Re-encryption	Client --TLS--> LB --TLS--> target	Encryption on both segments; more certificate/application work
Pass-through	Client --TLS bytes through NLB--> target	Target owns TLS and sees the original TLS session

ACM, SNI and security policies

AWS Certificate Manager issues or imports certificates for compatible listeners. Server Name Indication allows one listener to select among certificates for several domains. A listener security policy defines allowed TLS versions and cipher suites; test client compatibility before enforcing stricter policies.

Mutual TLS

With mutual TLS, clients present certificates that can be validated against a trust store. It is useful for device, partner and workforce authentication. Plan certificate issuance, rotation, revocation, subject mapping and failure handling before enabling it.

ALB authentication actions

An HTTPS listener rule can redirect users through Amazon Cognito or an OIDC provider, then forward authenticated identity information to the target. JWT validation can reject invalid access tokens before the application receives a request. The application must still enforce authorization for the requested resource.

Identity boundary

Authentication establishes who the caller is. Authorization decides what that identity may do. Moving authentication to ALB does not remove application authorization requirements.

PART IV - SECURITY

17. WAF, DDoS Protection and Application Defense

AWS WAF with ALB

AWS WAF evaluates HTTP requests against a web ACL. Rules can match IP addresses, countries, headers, paths, request rates and known exploit patterns. Managed rule groups accelerate coverage but still require tuning for the application's legitimate traffic.

Rate-based controls

Rate-based WAF rules can limit abusive request sources before requests reach targets. Application-level quotas may still be needed because business identities can share addresses and malicious traffic can be distributed across many sources.

AWS Shield

Shield Standard provides baseline DDoS protection for supported AWS services. Shield Advanced adds expanded detection, response assistance and cost-protection capabilities for enrolled resources. DDoS defense also depends on scalable architecture, caching and controlled origin exposure.

Defense in depth



Each layer handles a different problem. Shield addresses volumetric network attacks, WAF filters HTTP patterns, ALB authentication validates identity and the application enforces domain-specific authorization and data rules.

Do not forget the origin

If targets are reachable directly from the internet, attackers may bypass WAF and ALB controls. Restrict target security groups so normal traffic must traverse the intended entry point.

PART V - OPERATIONS

18. Monitoring, Logs and Operational Signals

Golden signals

Signal	Representative ELB evidence	Question answered
Traffic	RequestCount, connections, processed bytes	How much work is arriving?
Errors	ELB 4xx/5xx and target 4xx/5xx	Who generated the failure?
Latency	TargetResponseTime and application tracing	Where is time being spent?
Saturation	Rejected connections, capacity units, target resources	Is a limit being approached?
Availability	HealthyHostCount, UnHealthyHostCount, zonal health	Can traffic be served?

Log types

- ALB access logs record request routing, target choice, response codes and timing fields.
- ALB connection logs and health-check logs provide connection and probe diagnostics where enabled.
- NLB access logs focus on TLS listener connections; VPC Flow Logs expose network accept/reject information.
- CloudTrail records administrative API actions such as listener and target-group changes.

Alarm design

Alarm on user impact and loss of redundancy, not only total failure. A useful set includes target 5xx rate, p95/p99 latency, zero or low healthy-target count per zone, rejected connections and unexpected capacity-unit growth.

Correlation

During an incident, correlate the same time window across CloudWatch metrics, access logs, VPC Flow Logs, target logs and deployment events. A single graph rarely identifies root cause.

Metric distinction

HTTPCode_ELB_5XX_Count points toward load-balancer-generated failures. HTTPCode_Target_5XX_Count means the target returned the 5xx. The remediation paths are different.

PART V - OPERATIONS

19. Troubleshooting Playbook

Use the traffic path

- 1 Resolve DNS and confirm the expected load balancer name and addresses.
- 2 Confirm the load balancer is active and the correct scheme/address type is used.
- 3 Verify the listener, certificate and security policy.
- 4 Verify security groups, NACLs and route tables in both directions.
- 5 Check rule priority and prove which rule matched.
- 6 Inspect target registration, enabled zones and health reason codes.
- 7 Connect directly from an allowed test host to the target port and health path.
- 8 Compare access-log timing and status fields with application logs.

Common ALB response codes

Code	Likely direction	Investigation
400	Malformed client request	Headers, request syntax and size limits
401/403	Authentication, WAF or application	Rule actions, identity provider, web ACL and target logs
502	Bad target response/reset	Keep-alive mismatch, TLS to target, invalid response
503	No usable service capacity	Registered/healthy targets, zonal capacity and rule destination
504	Target connect/response timeout	Routes, security, target latency and timeouts

Healthy but application fails

A target can pass /health and still fail a specific route because of dependency errors, host-header assumptions, authorization, payload size, database exhaustion or deployment defects. Health is one signal, not proof of business correctness.

Fast isolation

Test four points separately: client to listener, listener rule selection, load balancer to target port, and target application behavior. This converts a vague outage into a bounded segment.

PART V - OPERATIONS

20. Cost, Capacity and Quotas

Billing model

ALB, NLB and GWLB are generally billed for running time plus consumed capacity units. Standard data transfer and public IPv4 charges can also apply. Gateway Load Balancer endpoints are separate billable resources.

ALB LCU dimensions

Dimension	Measures	Optimization thought
New connections	Connection establishment rate	Reuse connections where appropriate
Active connections	Concurrent established connections	Understand WebSockets and long-lived clients
Processed bytes	Request and response data	Use caching/compression appropriately
Rule evaluations	Rules evaluated against request rate	Order and consolidate rules intentionally

The billable ALB capacity for a period is based on the highest normalized LCU dimension, not the sum of all four. NLB and GWLB use their own capacity-unit dimensions.

Capacity reservations

Capacity-unit reservations can establish a minimum prepared capacity for predictable high-volume events. They reduce scaling uncertainty but create reserved charges, so size them from measured traffic and load tests.

Quotas

Accounts have regional quotas for load balancers, target groups, listeners, rules, certificates and targets. Some are adjustable and others are fixed. Check Service Quotas early, especially before migrations, multi-tenant platforms and peak events.

Cost checklist

- Remove unused load balancers and target groups.
- Review public IPv4 and cross-zone architecture implications.
- Monitor consumed capacity units and processed bytes.
- Use CloudFront caching for cacheable public content when appropriate.
- Consolidate only where ownership and blast-radius tradeoffs remain acceptable.

PART VI - ARCHITECTURE AND AUTOMATION

21. Integrations with the AWS Platform

Service	Integration	Why it matters
Route 53	Alias records and health-based routing	Friendly names and regional failover
Auto Scaling	Registers/deregisters EC2 targets	Elastic compute capacity
ECS	Registers tasks in target groups	Service deployments and dynamic task addresses
EKS	AWS Load Balancer Controller	ALB Ingress and NLB Service patterns
CloudFront	Uses ALB as an origin	Edge caching and global delivery
Global Accelerator	Routes global anycast traffic to endpoints	Global static addresses and fast failover
PrivateLink	NLB-backed endpoint service	Private cross-VPC/account service consumption
WAF	Associates a web ACL with ALB	HTTP request filtering

ALB versus API Gateway

ALB is efficient for HTTP routing to services and containers. API Gateway is purpose-built for managed APIs and adds capabilities such as API keys, usage plans, request transformations, stages and direct AWS service integrations. Some architectures use API Gateway publicly and an internal load balancer privately.

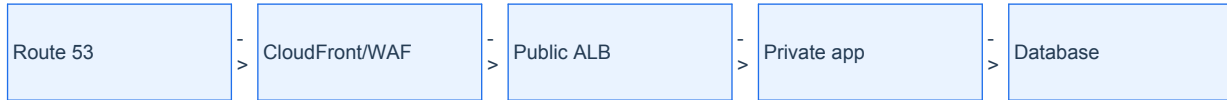
CloudFront versus Global Accelerator

CloudFront is an HTTP content delivery network with caching and edge behavior. Global Accelerator is a network traffic accelerator with anycast addresses and endpoint health routing. Choose based on protocol, caching needs and global traffic-control requirements.

PART VI - ARCHITECTURE AND AUTOMATION

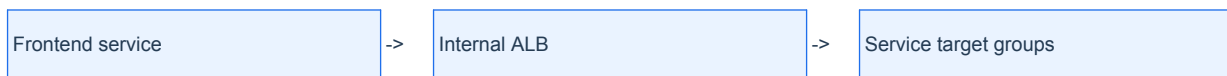
22. Architecture Patterns and Tradeoffs

Public three-tier application



Only the ALB is internet-facing. Application and database tiers use private subnets and tightly scoped security groups. CloudFront is optional but useful for caching and edge controls.

Internal microservices



An internal ALB can consolidate host/path routing for a bounded service domain. For very large service meshes, evaluate service discovery and mesh technologies rather than putting every east-west call behind one central ALB.

Static entry plus HTTP routing



Use this only when static zonal addresses or PrivateLink must coexist with ALB features. Monitor both layers and document which resource owns TLS termination.

Blue-green release

```

Listener forward action:
  blue target group = 90
  green target group = 10
  
```

Increase green only when technical and business metrics remain healthy.

Centralized appliance inspection



This centralizes security policy while increasing route-table and shared-service dependency. Design multi-AZ appliance capacity and a deliberate failure posture.

PART VI - ARCHITECTURE AND AUTOMATION

23. Infrastructure as Code and Change Safety

Why use IaC

Listeners, rules, target-group attributes and security controls form an interdependent system. Infrastructure as code makes those decisions reviewable, repeatable and recoverable across environments.

Resource ownership

- Define load balancer, listeners, target groups and security groups in the same ownership boundary where practical.
- Avoid two deployment systems modifying the same listener rules.
- Parameterize environment-specific names, certificates, subnets and domains without hiding architectural differences.
- Use tags for application, environment, owner, cost center and data classification.

Safe change sequence

- 1 Create or update the target group and register new capacity.
- 2 Wait for health and application readiness.
- 3 Add or modify routing with a testable priority.
- 4 Observe metrics and logs during traffic shift.
- 5 Drain old targets before removal.
- 6 Remove obsolete rules only after rollback is no longer required.

Example AWS CLI inspection commands

```
aws elbv2 describe-load-balancers
aws elbv2 describe-listeners --load-balancer-arn <arn>
aws elbv2 describe-rules --listener-arn <arn>
aws elbv2 describe-target-health --target-group-arn <arn>
aws elbv2 describe-target-group-attributes --target-group-arn <arn>
```

Automation principle

Automate creation and verification. A deployment is not complete merely because the API accepted the change; target health, rule behavior and user-facing signals must also be checked.

PART VII - PRACTICE

24. Hands-On Learning Labs

Lab 1 - Build a highly available ALB

- 1 Create or select a VPC with two public subnets and two private application subnets across two Availability Zones.
- 2 Run two simple HTTP targets in different zones and expose a /health endpoint.
- 3 Create an internet-facing ALB in the public subnets and a target group for the private targets.
- 4 Allow port 80 temporarily, verify routing, then add an ACM certificate and HTTPS listener.
- 5 Redirect HTTP to HTTPS and confirm targets cannot be reached directly from the internet.
- 6 Stop one target and observe health removal and continued service.

Success criteria

- HTTPS works through a friendly Route 53 name
- Only healthy targets receive requests
- Direct target access is blocked
- CloudWatch shows requests and target health

Lab 2 - Host and path routing

- 1 Create separate web, API and admin target groups.
- 2 Route /api/* to API, admin.example.com to admin and all remaining requests to web.
- 3 Add a fixed 403 response for an intentionally blocked path.
- 4 Test positive, negative and overlapping rule cases.

Lab 3 - Canary deployment

- 1 Deploy version 1 and version 2 to separate target groups.
- 2 Forward 95 percent to v1 and 5 percent to v2.
- 3 Compare latency, 5xx and a business-result metric.
- 4 Increase gradually or roll back to 100 percent v1.

Lab 4 - NLB client IP and TLS

- 1 Create an NLB with targets in two zones.
- 2 Observe the zonal static addresses and target-side source address.
- 3 Compare TLS termination at NLB with TCP pass-through to the target.
- 4 Test deregistration while a long-lived connection exists.

Lab 5 - Troubleshooting game day

Introduce one fault at a time: blocked target port, wrong health path, incorrect rule priority, expired certificate, slow target and empty target group. Diagnose using the traffic-path checklist without immediately viewing the injected fault. Use small temporary resources, set a budget alert and delete labs when finished.

PART VII - PRACTICE

25. Production Readiness Checklist

Area	Questions to answer before launch
Architecture	Correct LB type? Multi-AZ? Enough surviving-zone capacity? Dependencies redundant?
Routing	Rules tested for matches and non-matches? Default rule safe? Redirects correct?
Targets	Health endpoint meaningful? Drain and startup behavior tested? Capacity autoscaling configured?
Network	Subnets have free addresses? Routes, SGs, NACLs and client IP behavior verified?
TLS	Certificates valid and renewable? SNI domains complete? Security policy compatible?
Security	Targets private? WAF tuned? Authentication and authorization boundaries documented?
Operations	Metrics, logs, alarms, dashboards and runbooks ready? Ownership and escalation known?
Resilience	Instance and zone failures tested? Regional recovery objective documented?
Cost	Expected LCUs/NLCUs, data transfer, endpoints and public IPv4 costs estimated?
Change	IaC reviewed? Rollback tested? Quotas checked? Maintenance communication ready?

Minimum useful alarms

- Healthy target count below the redundancy requirement.
- Target 5xx rate above the service objective.
- ELB 5xx count above baseline.
- p95/p99 target response time degradation.
- Rejected connections or unexpected capacity-unit growth.
- Certificate expiration and configuration-change notifications.

PART VII - PRACTICE

26. Interview and Revision Questions

Check your understanding

- 1 Explain the difference between ALB, NLB and GWLB without using only the OSI layer numbers.
- 2 Why can an internet-facing ALB safely route to private targets?
- 3 When would you deliberately place an NLB in front of an ALB?
- 4 How can stickiness cause uneven traffic, and how would you redesign the application to avoid it?
- 5 Why can every ALB target be healthy while users still receive errors?
- 6 Compare cross-zone enabled and disabled behavior during a zone failure.
- 7 How would you troubleshoot an ALB 502 differently from a target-generated 500?
- 8 What is the relationship between deregistration delay and container termination grace?
- 9 How do WAF, Shield, ALB authentication and application authorization differ?
- 10 Why does an NLB static IP requirement not automatically mean the application should use only an NLB?
- 11 Which ALB LCU dimension would a WebSocket-heavy application likely watch closely, and why?
- 12 Design a safe blue-green rollout and define its rollback signals.

Scenario challenge

Design prompt

A partner requires two fixed public IP addresses. The service needs path routing, JWT validation and WAF. Propose an architecture, explain the traffic path, identify TLS ownership and describe the additional cost and failure surface.

APPENDICES

27. Quick Reference

One-page selector

Requirement	Default choice	Reason
Host/path/header routing	ALB	Understands HTTP request content
gRPC application routing	ALB	Supports gRPC-aware target protocol
Static zonal addresses	NLB	Stable address per enabled zone
UDP or QUIC	NLB	Transport-level protocol support
Private endpoint service	NLB	Backs AWS PrivateLink endpoint services
Virtual firewall fleet	GWLB	Transparent appliance insertion
Legacy ELB	Migrate CLB	Modern types provide richer capabilities

Troubleshooting mnemonic

DNS -> Scheme/IP -> Listener/TLS -> Security/Routes -> Rule -> Target group -> Health -> Application -> Logs

Core resource relationships

```
Load balancer
  listener (protocol + port + certificate/policy)
    ordered rules (conditions + actions)
      target group (protocol + port + health + attributes)
        registered targets
```

Production defaults to question

- Two or more Availability Zones - but confirm per-zone target capacity.
- Private application targets - but confirm required egress and management access.
- HTTPS at the edge - but decide whether the back-end segment also needs TLS.
- Stateless targets - but define shared session and data stores.
- Health checks - but design what healthy actually means.
- Logs and alarms - but connect them to an owned incident response process.

APPENDICES

28. Glossary

Term	Meaning
ACM	AWS Certificate Manager; manages compatible TLS certificates.
ALB	Application Load Balancer; HTTP-aware Layer 7 load balancer.
Availability Zone	An isolated location within an AWS Region.
Cross-zone	Routing from a load-balancer node to targets in other enabled zones.
Deregistration delay	Time allowed for existing work to drain after target removal begins.
ELB	Elastic Load Balancing; the AWS load-balancing service family.
GENEVE	Encapsulation protocol used between GWLB and compatible appliances on port 6081.
GWLB	Gateway Load Balancer; distributes traffic to virtual network appliances.
Health check	Periodic probe used to determine whether a target should receive traffic.
Listener	Protocol and port on which a load balancer accepts connections.
NLB	Network Load Balancer; transport/flow-oriented load balancer.
OIDC	OpenID Connect; identity layer used by ALB user authentication.
PrivateLink	Private service connectivity using VPC endpoints without public routing.
Proxy Protocol v2	Binary header carrying original connection metadata to compatible targets.
Rule	Conditions and actions that determine how matching listener traffic is handled.
SNI	TLS extension allowing selection among certificates for multiple hostnames.
Stickiness	Affinity that prefers the same target for a client or flow.
Target	Destination that performs application or appliance work.
Target group	Pool of targets plus routing, health and lifecycle policy.
WAF	Web application firewall that evaluates HTTP requests against rules.

APPENDICES

29. Official AWS References

AWS features, quotas, protocols and prices evolve. Verify implementation details in current regional documentation before production changes.

[Elastic Load Balancing documentation](https://docs.aws.amazon.com/elasticloadbalancing/)

<https://docs.aws.amazon.com/elasticloadbalancing/>

[How Elastic Load Balancing works](https://docs.aws.amazon.com/elasticloadbalancing/latest/userguide/how-elastic-load-balancing-works.html)

<https://docs.aws.amazon.com/elasticloadbalancing/latest/userguide/how-elastic-load-balancing-works.html>

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[ALB target groups](https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-target-groups.html)

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<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/rule-action-types.html>

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[NLB target-group attributes](https://docs.aws.amazon.com/elasticloadbalancing/latest/network/edit-target-group-attributes.html)

<https://docs.aws.amazon.com/elasticloadbalancing/latest/network/edit-target-group-attributes.html>

[Gateway Load Balancers](https://docs.aws.amazon.com/elasticloadbalancing/latest/gateway/introduction.html)

<https://docs.aws.amazon.com/elasticloadbalancing/latest/gateway/introduction.html>

[ALB CloudWatch metrics](https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-cloudwatch-metrics.html)

<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-cloudwatch-metrics.html>

[ALB access logs](https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-access-logs.html)

<https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-access-logs.html>

[Elastic Load Balancing pricing](https://aws.amazon.com/elasticloadbalancing/pricing/)

<https://aws.amazon.com/elasticloadbalancing/pricing/>

Document scope

This guide is an educational architecture and operations reference. It does not replace workload-specific security review, load testing, quota verification or current AWS documentation.